COURSE 2 · ALGEBRA

# Algebra

*Printable flashcards*

143 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

# Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

## Module 2.1 · Expressions and simplifying

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| EQUATION<br>Write the equation: Distributive law | $a(b+c) = ab + ac$ |
| EQUATION<br>Write the equation: Expanding two brackets | $(a+b)(c+d) = ac + ad + bc + bd$ |
| EQUATION<br>Write the equation: Index laws | $x^a x^b = x^{a+b}, \quad \frac{x^a}{x^b} = x^{a-b}, \quad (x^a)^b = x^{ab}$ |
| EQUATION<br>Write the equation: Negative and fractional indices | $x^{-n} = \frac{1}{x^n}, \qquad x^{1/n} = \sqrt[n]{x}$ |
| EQUATION<br>Write the equation: Difference of two squares | $a^2 - b^2 = (a-b)(a+b)$ |
| KEY IDEA<br>Complete the key idea: Only like … | Only like terms combine |
| KEY IDEA<br>Complete the key idea: Expanding and factorising are … | Expanding and factorising are inverses: check one with the other |
| KEY IDEA<br>Complete the key idea: $x^a x^b = x^{a+b}$; $(x^a)^b = x^{ab}$; … | $x^a x^b = x^{a+b}$; $(x^a)^b = x^{ab}$; $x^{-n} = 1/x^n$; x^(1/n) is the nth root |
| QUICK CHECK<br>3x and $3x^2$ are | unlike terms |
| QUICK CHECK<br>$x^{-2}$ equals | $1/x^2$ |

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| QUICK CHECK<br>$x^2 + 5x + 6$ factorises as | $(x + 2)(x + 3)$ |
| QUICK CHECK<br>x^(1/2) means | $\sqrt{x}$ |
| PROBLEM<br>Simplify $4x + 3y - x + 2y$. | $3x + 5y$. Collect y terms: $3y + 2y = 5y$. |
| PROBLEM<br>Expand $5(2x - 3)$. | $10x - 15$. Multiply each term by 5. |
| PROBLEM<br>Expand $(x + 4)(x - 2)$. | $x^2 + 2x - 8$. $x^2 + 2x - 8$. |
| PROBLEM<br>Factorise $x^2 - 7x + 12$. | $(x - 3)(x - 4)$. Two numbers with sum $-7$ and product 12: $-3$ and $-4$. |
| PROBLEM<br>Factorise $9x^2 - 25$. | $(3x - 5)(3x + 5)$. Difference of two squares: $(3x)^2 - 5^2$. |
| PROBLEM<br>Evaluate 8^(2/3). | 4. $2^2 = 4$. |
| PROBLEM<br>Simplify $(x^3 \times x^5) \div x^2$. | $x^6$. Divide: subtract to get $x^6$. |

## Module 2.2 · Linear equations and inequalities

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| EQUATION<br>Write the equation: Straight line | $y = mx + c$ |
| EQUATION<br>Write the equation: Inequality sign flip | $-x > 3 \Rightarrow x < -3$ |
| EQUATION<br>Write the equation: Gradient between two points | $m = \frac{y_2 - y_1}{x_2 - x_1}$ |
| KEY IDEA<br>Complete the key idea: Do the same to … | Do the same to both sides; check by substitution |

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| KEY IDEA<br>Complete the key idea: Multiplying or dividing an inequality by a negative … | Multiplying or dividing an inequality by a negative flips its direction |
| KEY IDEA<br>Complete the key idea: Gradient = rise ÷ run; … | Gradient = rise ÷ run; in science it usually means something physical |
| QUICK CHECK<br>Solve $2x = 18 - x$. | $x = 6$ |
| QUICK CHECK<br>Dividing an inequality by −3 | reverses its direction |
| QUICK CHECK<br>In $y = 3x - 5$, the intercept is | −5 |
| QUICK CHECK<br>A line with gradient 0 is | horizontal |
| PROBLEM<br>Solve $4x + 7 = 31$. | 6. Divide by 4: $x = 6$. |
| PROBLEM<br>Solve $3(x - 2) = 15$. | 7. $3x = 21$, $x = 7$. |
| PROBLEM<br>Solve $x/4 + 2 = x/2 - 1$. | 12. $12 = x$. |
| PROBLEM<br>Solve $5 - 2x > 11$. | $x < -3$. Divide by −2 and flip: $x < -3$. |
| PROBLEM<br>Find the gradient of the line through (1, 3) and (5, 11). | 2. = 2. |
| PROBLEM<br>A taxi charges a $4 flag fall plus $2.50 per km. How far can you go for $29? | 10 km. $2.5d = 25$, $d = 10$ km. |
| PROBLEM<br>On a graph of distance (m) against time (s), the gradient represents | speed. Gradient = change in distance ÷ change in time = speed. |

## Module 2.3 · Rearranging formulas

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| EQUATION<br>Write the equation: Simple pendulum | $T = 2\pi\sqrt{\frac{L}{g}}$ |
| EQUATION<br>Write the equation: Rearranged for g | $g = \frac{4\pi^2 L}{T^2}$ |
| EQUATION<br>Write the equation: Thin lens, rearranged for v | $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow v = \frac{uf}{u-f}$ |
| KEY IDEA<br>Complete the key idea: Undo operations in reverse … | Undo operations in reverse order, applied to both whole sides |
| KEY IDEA<br>Complete the key idea: Flip only whole … | Flip only whole sides, never individual terms |
| KEY IDEA<br>Complete the key idea: Target twice: … | Target twice: collect, factor, divide |
| QUICK CHECK<br>To solve T = 2π√(L/g) for g, the first step is | divide by 2π |
| QUICK CHECK<br>If the target appears twice, you should | collect and factor |
| QUICK CHECK<br>v = √(2E/m). If E is quadrupled, v | doubles |
| QUICK CHECK<br>Taking the reciprocal is allowed | of a whole side only |
| PROBLEM<br>A pendulum clock must have a period of 2.00 s where g = 9.81 m/s$^2$. How long is the pendulum? | 0.994 m. = 9.81 × 4 ÷ 39.48 ≈ 0.994 m. |
| PROBLEM<br>From 1/f = 1/u + 1/v, a student writes f = u + v. The mistake is | flipping individual terms. The reciprocal of a sum is not the sum of reciprocals: 1/2 + 1/2 = 1, but 2 + 2 ≠ 1. |

## Module 2.4 · Functions and graphs

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| EQUATION<br>Write the equation: General transformation | $y = a\,f(b(x-h)) + k$ |
| EQUATION<br>Write the equation: Vertex form of a parabola | $y = a(x-h)^2 + k$ |
| EQUATION<br>Write the equation: Composite and inverse | $(f \circ g)(x) = f(g(x)), \qquad f^{-1}(f(x)) = x$ |
| KEY IDEA<br>Complete the key idea: A function gives … | A function gives exactly one output per input |
| KEY IDEA<br>Complete the key idea: Domain: allowed … | Domain: allowed inputs. Range: resulting outputs |
| KEY IDEA<br>Complete the key idea: f(x − h) + k shifts … | f(x − h) + k shifts right h and up k; −f(x) reflects in the x-axis |
| QUICK CHECK<br>A function can map one input to | exactly one output |
| QUICK CHECK<br>y = f(x) + 3 is shifted | up 3 |
| QUICK CHECK<br>y = −f(x) is | reflected in the x-axis |
| QUICK CHECK<br>The domain of 1/x excludes | x = 0 |
| PROBLEM<br>If f(x) = 3x² − 2, find f(4). | 46. = 46. |
| PROBLEM<br>What is the domain of f(x) = √(x − 5)? | x ≥ 5. The quantity under the root must be ≥ 0: x − 5 ≥ 0. |
| PROBLEM<br>The graph of y = (x + 3)² is the graph of y = x² shifted | 3 left. x − h with h = −3: shift left by 3. |

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| PROBLEM<br>The vertex of $y = 2(x - 1)^2 + 5$ is at | (1, 5). Vertex form $a(x - h)^2 + k$ has vertex (h, k). |
| PROBLEM<br>If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$. | 19. $f(9) = 19$. |
| PROBLEM<br>What is the inverse of $f(x) = (x - 4)/3$? | $3x + 4$. So $f^{-1}(x) = 3x + 4$. |
| PROBLEM<br>Which function has a vertical asymptote at $x = 2$? | $y = 1/(x - 2)$. Division by zero at $x = 2$. |
| PROBLEM<br>Sketch in words how $y = -2(x - 3)^2 + 4$ relates to $y = x^2$. | Shift right 3, stretch vertically by a factor of 2, reflect upside down, then shift up 4. It is an upside-down parabola, narrower than $y = x^2$, with its highest point at (3, 4). |

## Module 2.5 · Systems of equations

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| EQUATION<br>Write the equation: General two-variable system | $a_1x + b_1y = c_1, \qquad a_2x + b_2y = c_2$ |
| EQUATION<br>Write the equation: Elimination step | $(2x + 3y) - (2x - y) = 12 - 4 \;\Rightarrow\; 4y = 8$ |
| KEY IDEA<br>Complete the key idea: Substitute when one … | Substitute when one variable is easy to isolate |
| KEY IDEA<br>Complete the key idea: Eliminate by … | Eliminate by matching coefficients and subtracting |
| KEY IDEA<br>Complete the key idea: Solution = intersection; … | Solution = intersection; parallel lines mean no solution |
| QUICK CHECK<br>Elimination works by | cancelling one variable |
| QUICK CHECK<br>Parallel lines give | no solution |
| QUICK CHECK<br>Three unknowns generally need | three equations |

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| QUICK CHECK<br>Solve x = 2y and x + y = 9: y = | 3 |
| PROBLEM<br>Solve x + y = 10 and x − y = 4. | x = 7, y = 3. y = 3. |
| PROBLEM<br>Solve 3x + 2y = 16 and x − y = 2. What is x? | 4. 3(y + 2) + 2y = 16 → 5y = 10 → y = 2, x = 4. |
| PROBLEM<br>Solve 2x + 5y = 19 and 3x − 2y = 0. What is y? | 3. Subtract: 19y = 57, y = 3. |
| PROBLEM<br>Adult tickets cost $15 and child tickets $8. 40 tickets raised $467. How many adult tickets were sold? | 21. 15a + 8(40 − a) = 467 → 7a = 147 → a = 21. |
| PROBLEM<br>The system y = 2x + 1 and y = 2x − 3 has | no solution. Same gradient, different intercepts: parallel lines. |
| PROBLEM<br>How many litres of 10% salt solution must be mixed with 30% solution to make 20 L of 25% solution? | 5 L. 0.1x + 0.3(20 − x) = 5 → −0.2x = −1 → x = 5 L. |
| PROBLEM<br>Explain geometrically why two linear equations in two unknowns can have exactly one, none, or infinitely many solutions, but never exactly two. | Each equation is a straight line. Two distinct straight lines either cross at exactly one point or are parallel and never meet. If they are the same line, every point is shared. Two straight lines cannot cross twice. |

## Module 2.6 · Quadratics and polynomials

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| EQUATION<br>Write the equation: Quadratic formula | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ |
| EQUATION<br>Write the equation: Discriminant | $\Delta = b^2 - 4ac$ |
| EQUATION<br>Write the equation: Factor theorem | $p(k) = 0 \iff (x - k) \text{ is a factor}$ |

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| EQUATION<br>Write the equation: Vertex of $y = ax^2 + bx + c$ | $x_{\text{vertex}} = -\frac{b}{2a}$ |
| KEY IDEA<br>Complete the key idea: Quadratic formula … | Quadratic formula solves every quadratic |
| KEY IDEA<br>Complete the key idea: Discriminant > 0: two … | Discriminant > 0: two roots; = 0: one; < 0: none |
| KEY IDEA<br>Complete the key idea: If p(k) = 0 … | If p(k) = 0 then (x − k) is a factor |
| QUICK CHECK<br>If $b^2 - 4ac < 0$, the quadratic has | no real roots |
| QUICK CHECK<br>A cubic has at most how many real roots? | 3 |
| QUICK CHECK<br>The vertex of $y = ax^2 + bx + c$ is at x = | −b/2a |
| QUICK CHECK<br>$x^2 - 4 = 0$ has roots | ±2 |
| PROBLEM<br>Solve $x^2 - 9x + 20 = 0$. | 4 and 5. Factorise: $(x - 4)(x - 5) = 0$. |
| PROBLEM<br>Use the quadratic formula to find the larger root of $2x^2 - 3x - 5 = 0$. | 2.5. x = (3 ± 7)/4: 2.5 or −1. |
| PROBLEM<br>Find the discriminant of $3x^2 + 4x + 2$. | -8. = −8, so no real roots. |
| PROBLEM<br>Find the x-coordinate of the vertex of $y = x^2 - 6x + 11$. | 3. = 3. |
| PROBLEM<br>A ball's height is $h = 20t - 5t^2$ metres. After how many seconds does it land? | 4 s. t = 0 (launch) or t = 4 s. |

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| PROBLEM<br>For the same ball, h = 20t − 5t², what is the maximum height? | 20 m. h = 40 − 20 = 20 m. |
| PROBLEM<br>p(x) = x³ − 6x² + 11x − 6 and p(1) = 0. Which is a factor? | x − 1. Factor theorem: p(1) = 0 means (x − 1) is a factor. |
| PROBLEM<br>For which value of k does x² + kx + 9 = 0 have exactly one root (k > 0)? | 6. k = 6. |

## Module 2.7 · Exponentials and logarithms

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| EQUATION<br>Write the equation: Discrete growth | $N = N_0(1+r)^t$ |
| EQUATION<br>Write the equation: Continuous growth and decay | $N = N_0 e^{kt}$ |
| EQUATION<br>Write the equation: Log laws | $\log(ab) = \log a + \log b, \quad \log \frac{a}{b} = \log a - \log b, \quad \log a^n = n \log a$ |
| EQUATION<br>Write the equation: Half-life | $t_{1/2} = \frac{\ln 2}{k}$ |
| EQUATION<br>Write the equation: Solving an exponential | $a^x = b \Rightarrow x = \frac{\ln b}{\ln a}$ |
| KEY IDEA<br>Complete the key idea: Exponential: multiply by … | Exponential: multiply by the same factor each step |
| KEY IDEA<br>Complete the key idea: Logs undo exponentials; … | Logs undo exponentials; log(aᵇ) = b log a |
| KEY IDEA<br>Complete the key idea: Half-life and doubling time are … | Half-life and doubling time are constant; t½ = ln 2 / k |
| KEY IDEA<br>Complete the key idea: Log scales: … | Log scales: +1 means ×10 |

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| QUICK CHECK<br>log(ab) equals | log a + log b |
| QUICK CHECK<br>After 2 half-lives, the fraction remaining is | ¼ |
| QUICK CHECK<br>ln is the logarithm to base | e |
| QUICK CHECK<br>On a log scale, an increase of 2 means | ×100 |
| PROBLEM<br>Evaluate $\log_{10} 10\,000$. | 4. $10^4 = 10\,000$. |
| PROBLEM<br>A sample has a half-life of 6 hours. What fraction remains after 18 hours? Give a decimal. | 0.125. $(½)^3 = 0.125$. |
| PROBLEM<br>Solve $2^x = 50$, to 3 significant figures. | 5.644. = 3.912 ÷ 0.693 ≈ 5.64. |
| PROBLEM<br>$1000 is invested at 5% a year compounded annually. What is it worth after 10 years? | 1629 dollars. ≈ $1629. |
| PROBLEM<br>Using the rule of 70, estimate the doubling time for growth of 7% a year. | 10 years. 70 ÷ 7 = 10 years. |
| PROBLEM<br>A substance decays with k = 0.1 per day. What is its half-life? | 6.931 days. 0.693 ÷ 0.1 ≈ 6.93 days. |
| PROBLEM<br>How many times more acidic is a solution of pH 2 than one of pH 5? | 1000 ×. $10^3 = 1000$. |
| PROBLEM<br>Simplify $\ln(e^5) + \log_{10}(100)$. | 7. 5 + 2 = 7. |

## Module 2.8 · Calculus intuition

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| EQUATION<br>Write the equation: Definition of the derivative | $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ |
| EQUATION<br>Write the equation: Power rule | $\frac{d}{dx} x^n = n x^{n-1}$ |
| EQUATION<br>Write the equation: Two special derivatives | $\frac{d}{dx} e^x = e^x, \qquad \frac{d}{dx} \sin x = \cos x$ |
| EQUATION<br>Write the equation: Area under a curve | $\int_a^b f(x)\, dx$ |
| EQUATION<br>Write the equation: Integrating a power | $\int x^n\, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$ |
| KEY IDEA<br>Complete the key idea: Derivative = instantaneous rate … | Derivative = instantaneous rate of change = gradient of the tangent |
| KEY IDEA<br>Complete the key idea: Power rule: … | Power rule: $d/dx(x^n) = nx^{n-1}$ |
| KEY IDEA<br>Complete the key idea: Derivative zero … | Derivative zero at peaks and troughs |
| KEY IDEA<br>Complete the key idea: Integral = area under … | Integral = area under the curve = accumulated total |
| QUICK CHECK<br>The derivative of a constant is | 0 |
| QUICK CHECK<br>Velocity is the derivative of | position |
| QUICK CHECK<br>The area under a speed-time graph gives | distance |
| QUICK CHECK<br>$d/dx(e^x) =$ | $e^x$ |

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| PROBLEM<br>Differentiate $y = x^4$. | $4x^3$. Bring down 4, reduce power to 3. |
| PROBLEM<br>Differentiate $y = 3x^2 + 5x - 7$. | $6x + 5$. $3 \times 2x + 5 + 0$. |
| PROBLEM<br>For $y = x^3 - 2x$, find the gradient at $x = 2$. | 10. At $x = 2$: $12 - 2 = 10$. |
| PROBLEM<br>A ball's height is $h = 30t - 5t^2$. What is its velocity, $dh/dt$, at $t = 1$ s? | 20 m/s. At $t = 1$: 20 m/s. |
| PROBLEM<br>For $h = 30t - 5t^2$, at what time is the ball at its highest? | 3 s. $t = 3$ s. |
| PROBLEM<br>A car's speed rises steadily from 0 to 20 m/s in 10 s. How far does it travel in that time (area under the graph)? | 100 m. $\frac{1}{2} \times 10 \times 20 = 100$ m. |
| PROBLEM<br>Evaluate the integral of $2x$ from $x = 0$ to $x = 3$. | 9. $3^2 - 0^2 = 9$. |
| PROBLEM<br>Explain in words what $dV/dt = -0.5$ litres per minute means for a draining tank. | The volume of water V is changing over time t at a rate of −0.5 litres per minute: the tank is losing half a litre every minute at that moment. The negative sign means volume is decreasing. |